

POPULAR IoT OPERATING SYSTEMS: The Role They Play

IoT operating systems have emerged as the backbone of the Internet of Things, enabling the seamless integration of billions of devices into a connected ecosystem. Selecting the right OS is essential.

The author, **SUFYAN BIN UZAYR**, apart from being a writer, is an educator and entrepreneur with multiple degrees in management, information technology, and humanities. He has written seven books and is director of a multinational IT company. As an educator, he has taught at universities and institutions worldwide.

The Internet of Things (IoT) refers to the interconnected network of devices embedded with sensors, actuators, and connectivity capabilities. These devices can collect, process, and exchange data with each other and with larger systems with the help of specially designed operating systems.

From smart homes that automate lighting and temperature control to industrial automation systems that optimise production processes, IoT devices are becoming increasingly ubiquitous, seamlessly weaving themselves into the fabric

of our daily lives. At the heart of these devices lie operating systems specifically designed for resource-constrained environments, known as IoT operating systems. These specialised OSes are tailored to handle the unique demands of connected devices, such as limited processing power, memory, and battery life. They must be efficient, secure, and capable of operating reliably in often harsh environments.

Role of IoT operating systems and their unique features

Operating systems play a crucial role in enabling the functionality of IoT devices. They provide the core software infrastructure that manages device resources, executes applications, and facilitates communication with other devices and networks. IoT operating systems are specifically designed to address the unique challenges of resource-constrained devices, such as limited processing power, memory, and battery life.

Unlike traditional operating systems, IoT operating systems are tailored to handle the unique demands of connected devices. They must be efficient, secure, and capable of operating on devices with limited processing power, memory, and battery life. Key features that differentiate

